Solution to analysis in Home Assignment 1

Osama Al Sheikh Ali + Chalmers ID (osamas)

Analysis

In this report I will present my independent analysis of the questions related to home assignment 1. I have discussed the solution with Mubina Crnic, but I swear that the analysis written here are my own.

1 Properties of random variables

a) Let x be a scalar Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$. Using the definition of expected value (integral form), show that:

i)

$$E[x] = \mu.$$

We can use the definition of the Gaussian distribution where X has probability density function:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We have the following definition for the expected value of a random variable:

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

Inserting F(x) inside E[x]

$$E(X) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx$$

Using hint 1:

$$t = \frac{x-\mu}{2\sigma}$$

$$= \frac{\sqrt{2}\sigma}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \left(\sqrt{2}\sigma t + \mu\right) \exp(-t^2) dt$$

$$= \frac{1}{\sqrt{\pi}} \left(\sqrt{2}\sigma \int_{-\infty}^{\infty} t \exp(-t^2) dt + \mu \int_{-\infty}^{\infty} \exp(-t^2) dt\right)$$

Using hint 2:

$$\int_{-\infty}^{\infty} e^{-tx^2} dx = \sqrt{\frac{\pi}{\sqrt{t}}}$$

$$= \frac{1}{\sqrt{\pi}} \left(\sqrt{2}\sigma \left[-\frac{1}{2}\exp(-t^2)\right]_{-\infty}^{\infty} + \mu\sqrt{\pi}\right)$$

$$= \frac{\mu\sqrt{\pi}}{\pi}$$

$$= \mu$$

ii)

General way of expressing variance is the following

$$Var(X) = E[(x-\mu)^2] = \sigma^2$$

$$Var(X) = \int_{-\infty}^{\infty} x^2 f_X(x) dx - (E(X))^2$$

inserting the probability density equation of the Gaussian distribution

$$Var(X) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x^2 \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx - \mu^2$$

Using hint 1:

$$t = \frac{x-\mu}{2\sigma}$$

$$= \frac{\sqrt{2}\sigma}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \left(\sqrt{2}\sigma t + \mu\right)^2 \exp(-t^2) dt - \mu^2$$

$$= \frac{1}{\sqrt{\pi}} \left(2\sigma^2 \int_{-\infty}^{\infty} t^2 \exp(-t^2) dt + 2\sqrt{2}\sigma\mu \int_{-\infty}^{\infty} t \exp(-t^2) dt + \mu^2 \int_{-\infty}^{\infty} \exp(-t^2) dt\right) - \mu^2$$

$$= \frac{1}{\sqrt{\pi}} \left(2\sigma^2 \int_{-\infty}^{\infty} t^2 \exp(-t^2) dt + 2\sqrt{2}\sigma\mu \left[-\frac{1}{2} \exp(-t^2) \right]_{-\infty}^{\infty} + \mu^2 \sqrt{\pi} \right) - \mu^2$$

$$= \frac{1}{\sqrt{\pi}} \left(2\sigma^2 \int_{-\infty}^{\infty} t^2 \exp(-t^2) dt + 2\sqrt{2}\sigma\mu \cdot 0 \right) + \mu^2 - \mu^2$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \left(\int_{-\infty}^{\infty} t^2 \exp(-t^2) dt \right)$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \left(\left[-\frac{t}{2} \exp(-t^2) \right]_{-\infty}^{\infty} + \frac{1}{2} \int_{-\infty}^{\infty} \exp(-t^2) dt \right)$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \int_{-\infty}^{\infty} \exp(-t^2) dt$$

Using hint 2:

$$\int_{-\infty}^{\infty} e^{-tx^2} dx = \sqrt{\frac{\pi}{\sqrt{t}}}$$

we get the following

$$= \frac{2\sigma^2 \sqrt{\pi}}{2\sqrt{\pi}}$$

$$= \sigma^2$$

b) Let q be a multi-variate random variable with known probability density function p(q). Further, let z = Aq where A is a constant matrix.

i) Using the integral form of expectation, show that E[z] = AE[q].

To show that $E[z] = AE[q]$, where $E[\cdot]$ denotes the expectation operator, we can use the integral form of expectation.

General speaking, The expectation of a function $g(q)$ with respect to the probability density function $p(q)$ is given by the integral:

$$E[g(q)] = \int_{-\infty}^{\infty} g(q)p(q)\, dq$$

Given that $z = Aq$, we want to find $E[z]$, which is the expectation of z:

$$E[z] = E[Aq] = \int_{-\infty}^{\infty} Aq\, p(q)\, dq$$

Using linearity of the integral and the fact that A is a constant matrix, we can pull A out of the integral:

$$E[z] = A \underbrace{\int_{-\infty}^{\infty} q\, p(q)\, dq}_{E[q]}$$

Notice that the integral inside the expectation is the expectation of q, denoted as $E[q]$. Therefore, we have:

$$E[z] = AE[q]$$

This demonstrates that $E[z] = AE[q]$, as required.

ii) Using the result in i), show that Cov[z] = ACov[q]AT where Cov[·] is the covariance operator

To show that

$$\text{Cov}[z] = A\text{Cov}[q]A^T$$

where

$$\text{Cov}[\cdot]$$

denotes the covariance operator, we can use the definition of covariance and the result obtained in part (i).
The covariance of two random variables x and y is defined as:

$$\text{Cov}[x, y] = E[(x - E[x])(y - E[y])]$$

For a random vector $\mathbf{q}$, the covariance matrix $\text{Cov}[\mathbf{q}]$ is defined as:

$$\text{Cov}[\mathbf{q}] = E[(\mathbf{q} - E[\mathbf{q}])(\mathbf{q} - E[\mathbf{q}])^T]$$

Given that $z = A\mathbf{q}$, we want to find $\text{Cov}[z]$, the covariance matrix of z. Using the linearity of covariance, we have:

$$\text{Cov}[z] = \text{Cov}[A\mathbf{q}] = E[(A\mathbf{q} - E[A\mathbf{q}])(A\mathbf{q} - E[A\mathbf{q}])^T]$$

Expanding this expression, we get:

$$\text{Cov}[z] = E[(A\mathbf{q} - AE[\mathbf{q}])(A\mathbf{q} - AE[\mathbf{q}])^T]$$

Using the result from part (i), $E[z] = AE[\mathbf{q}]$, we can simplify this to:

$$\text{Cov}[z] = E[(A\mathbf{q} - AE[\mathbf{q}])(A\mathbf{q} - AE[\mathbf{q}])^T]$$

$$= E[(A\mathbf{q} - AE[\mathbf{q}])(\mathbf{q}^T A^T - E[\mathbf{q}]^T A^T)]$$

$$= E[A(\mathbf{q} - E[\mathbf{q}])(\mathbf{q} - E[\mathbf{q}])^T A^T]$$

$$= AE[(\mathbf{q} - E[\mathbf{q}])(\mathbf{q} - E[\mathbf{q}])^T]A^T$$

$$= A\text{Cov}[\mathbf{q}]A^T$$

Thus, we have shown that

$$\text{Cov}[z] = A\text{Cov}[\mathbf{q}]A^T$$

c)

We have the following probability

$$p(q) = N\left(\begin{bmatrix} 0 \\ 10 \end{bmatrix}, \begin{bmatrix} 0.3 & 0 \\ 0 & 8 \end{bmatrix}\right)$$

and now have the following transformation

$$z = Aq$$

where

$$A = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$$

from the beginning we have the following mean and covariance.

$$\mu_q = \begin{bmatrix} 0 \\ 10 \end{bmatrix}$$

$$\Sigma_q = \begin{bmatrix} 0.3000 & 0 \\ 0 & 8.0000 \end{bmatrix}$$

After the transformation and using the results in b) we get the following results

$$\mu_y = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$\Sigma_y = \begin{bmatrix} 2.3000 & 4.0000 \\ 4.0000 & 8.0000 \end{bmatrix}$$

and using the function sigmaEllipse2D we plot the level curve of the probability of the original probability and the transformed probability.

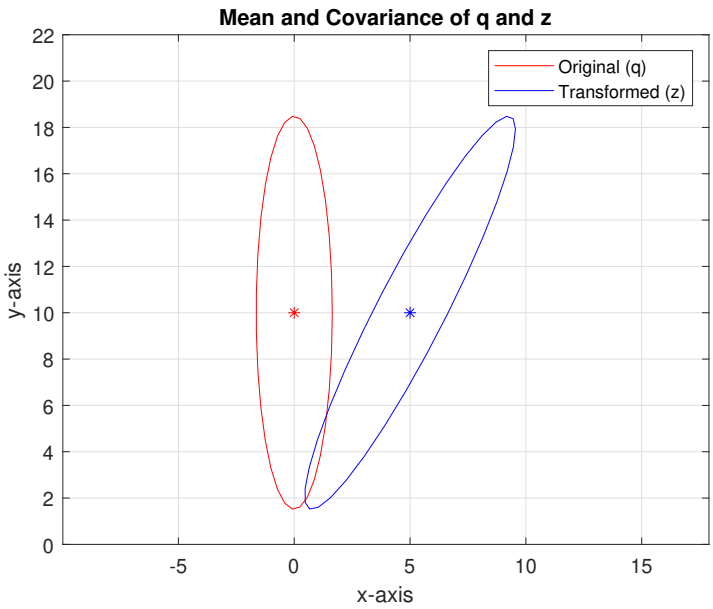

Figure 1.1: The level curve of the probability of the original probability and the transformed probability

What are the observations? We can say the A changes the mean of the original distribution to move more right as shown in the figure above. This means that mean went from being at (0,10) to (5,10) which is a higher mean in the x direction. For the covariance we can see that it became larger and this is visible from the graph as well where in the original case there was no correlation between x and y but after the transformation we have a positive correlation which means that if y increase so does x and vice versa. Looking at the individual values of covariance we can see that the correlation between different values of x increased from 0.3 to

2.3 while there was no change between in correlation between different values of y. However, look at the non diagonal elements we can see a change in value from 0 to 4. These values mean the correlation between x and y which totally match our conclusion from the graph. This can be traced back to the values of A. If A only had diagonal elements which were 1 then the transformation will still have the same mean and covariance but since there is 0.5 on the first row (non diagonal element) this will change them a bit. Since we only have 0.5 in the first row, it will only affect the mean of the x variable after the transformation which also matches our results. This 0.5 is also for the reason in the increase of correlation values in the covariance matrix which are related to x, but it will not lead to changes in correlation related to only the y values which is exactly why the correlation component related to y has the same value 8.

2 Properties of random variables

a)

To solve this problem, we first need to analytically determine the mean and variance of the transformed random variable $z = 3x$, where x is a Gaussian random variable with mean $\mu_x = 0$ and variance $\sigma_x^2 = 2$.

Given that $z = 3x$, we have:

$$E[z] = E[3x] = 3E[x] = 3 \times 0 = 0$$

$$\text{Var}[z] = \text{Var}[3x] = 3^2\text{Var}[x] = 3^2 \times 2 = 18$$

So, we have $z \sim N(0, 18)$.

Next, we'll use the provided function `approxGaussianTransform` to numerically approximate the mean and covariance of z. We'll compare the results by jointly plotting the histogram of the transformed samples along with Gaussian PDFs having the calculated and approximated mean and covariance. One observation that can be made is that the transformation Z will have a larger bandwidth than X and will be more flat.

The result is shown in the following graph for number of samples N = 5000

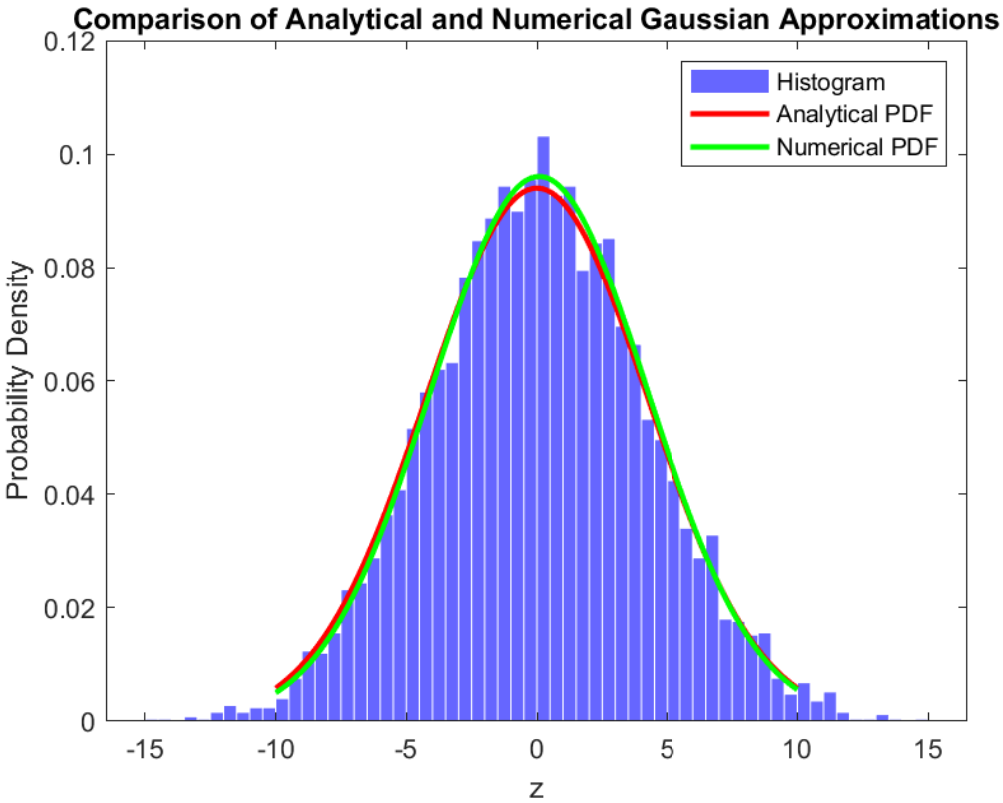

Figure 2.1: Analytical, Numerical and histogram N = 5000

and for N= 100000

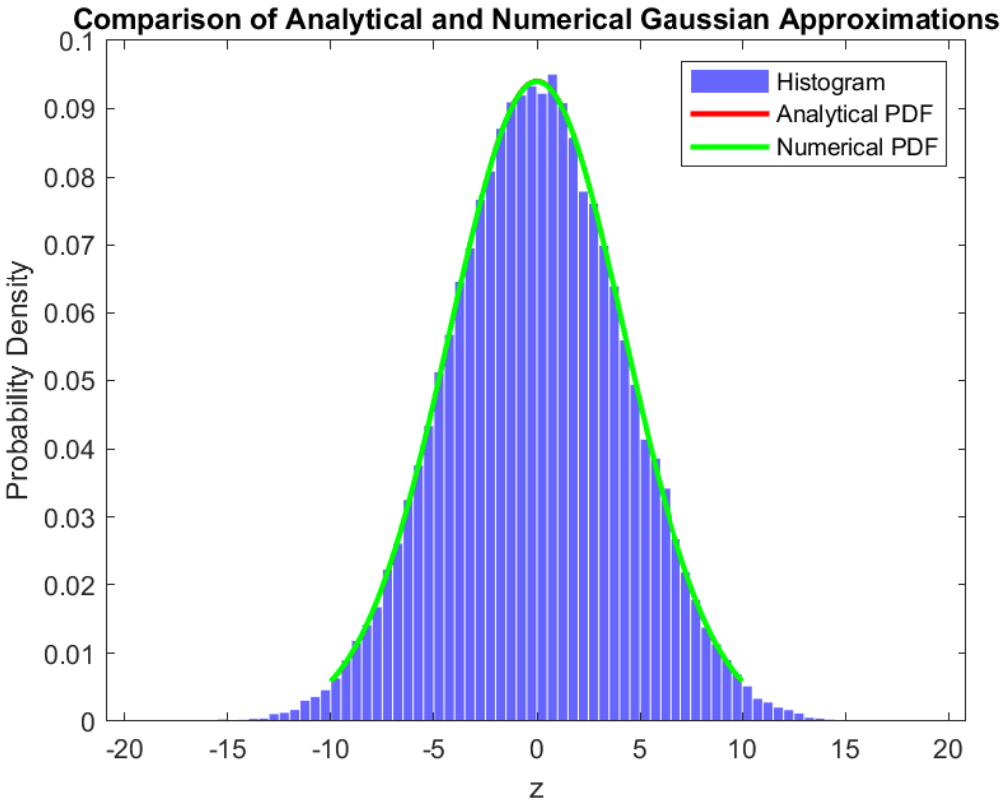

Figure 2.2: Analytical, Numerical and histogram N=100000

From the graphs, several observations can be made. Firstly, the analytical solution ("correct PDF") and the numerical solution ("estimation") match quite well, indicating the accuracy of our Gaussian distribution approximation. Additionally, the histogram of samples closely resembles a normal distribution shape, as evidenced by the graphs. It can be inferred that with a larger sample size, both the histogram and numerical estimations would further converge toward the analytical solution, as observed in the second graph. However, there appears to be a point of diminishing returns, where increasing the number of samples from 50,000 to 100,000 does not yield significant improvements. Instead, it only results in increased computational load. This suggests the existence of a cutoff point, indicating that continuously increasing the sample size may not always lead to better results.

b)

To solve part (b), we need to analyze the transformation of the random variable $z = x^3$.

Given that $z = x^3$, we want to determine $p(z)$ by assuming it is Gaussian and calculating $E[z]$ and $\mathrm{Var}[z]$ analytically.

However, it's not possible to directly express $p(z)$ as a Gaussian distribution because the transformation $z = x^3$ is nonlinear. Therefore, we cannot use the same approach as in part (a) where we analytically determined the mean and variance of the transformed random variable.

Instead, we can analyze the effect of this nonlinear transformation on the properties of $p(z)$ (histogram) and compare it to the Gaussian approximation provided by `approxGaussianTransform`. We can still plot the histogram of the transformed samples and the Gaussian PDFs with the calculated and approximated mean and covariance to visualize the differences between the true distribution and the approximation.

Since the transformation is nonlinear, the properties of $p(z)$ will not match the Gaussian approximation because it will not be Gaussian. This matches the results in the following

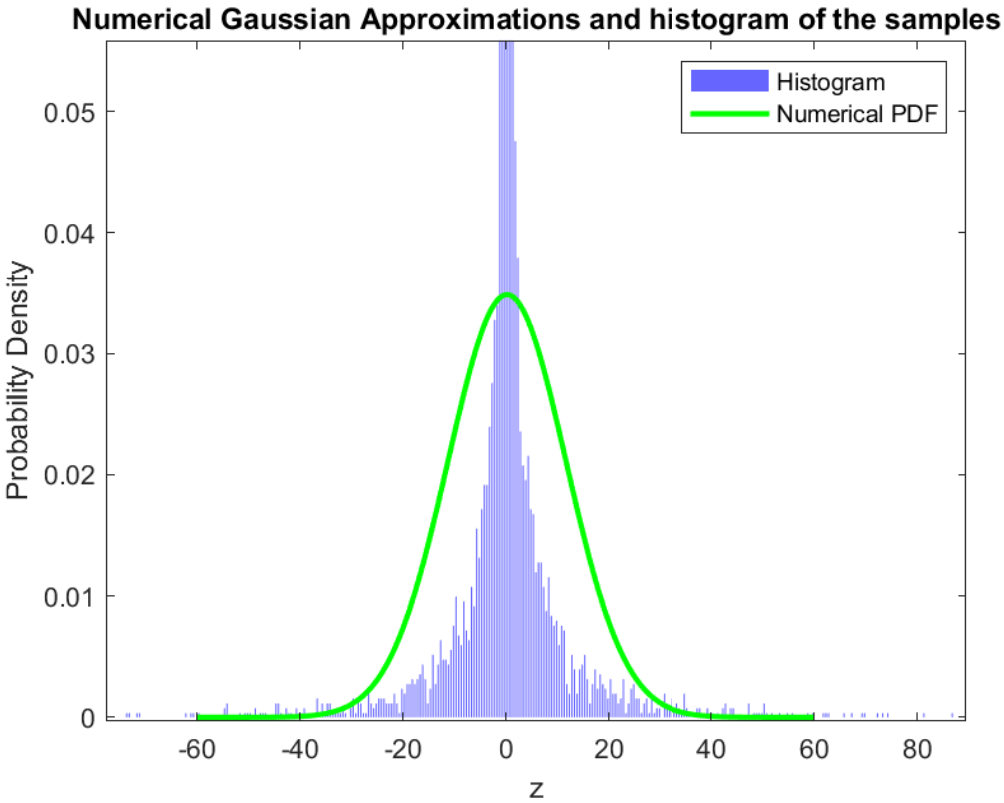

Figure 2.3: Numerical and histogram

and for N= 100000

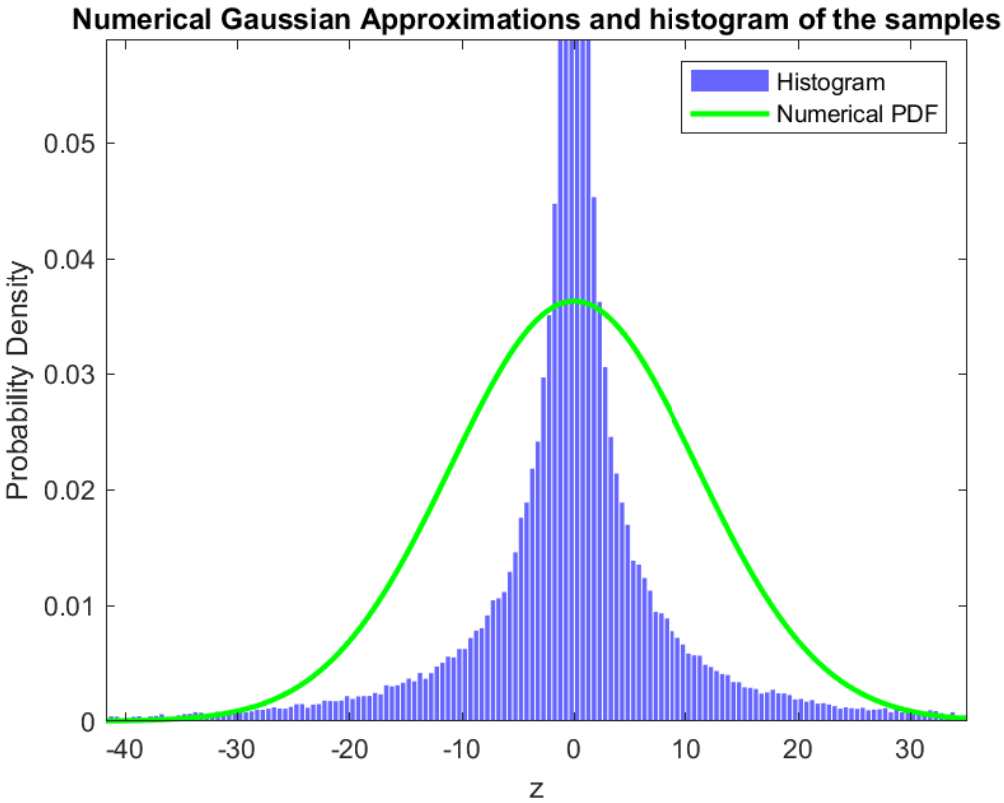

Figure 2.4: Numerical and histogram

c)

Now comparing b and a. As mentioned in b, we could not find an analytical solution to the problem since the transformation function is nonlinear. This will cause kurtosis and skewness. However, plotting the histogram and the numerical solution is still possible. What we can do is to compare these two together. We know that when we have a nonlinear transformation function just like in our case now, the resulting variable Z will no longer be Gaussian distributed. This can be investigated from the graphs when comparing histogram (the sampled values after transformation) with the numerical solution which is basically how a Gaussian distribution would look like given the new estimated mean and variance. What we observe is that the sampled values do not follow a normal Gaussian distribution at all as expected. Meanwhile in a, the transformation function is linear and the transformation will still exhibit normal Gaussian behaviour.

3 Understanding the conditional density

a)

When you add a normally distributed random variable (in our case the noise R) to a specific distribution h(x), the resulting distribution may or may not be a normal distribution. It depends on the nature of the specific distribution and the properties of the added normal variable.

In the scenario described, where $y = h(x) + r$ and r follows a normal distribution with mean 0 and variance σ_r^2, determining the distribution of y depends on the function $h(x)$ and its relationship with x.

If $h(x)$ is a linear transformation of x, then the distribution of y will also be the same as the distribution of x as can be seen in the graphs below where when h(x) was linear as in (5 * x) we can see that the output y was either uniformly or normally distributed depending on the distribution of x. (see d and e). So basically say x is uniformly distributed $U \sim (0, 10)$, y will also be uniformly distributed.

However, if $h(x)$ is a nonlinear function of x, the distribution of y may not be easily determined. The resulting distribution will depend on the specific form of $h(x)$ and its interaction with the normal noise term r which is very hard to determine.

In general, if $h(x)$ introduces significant non linearity or complexity, the distribution of y may not have a simple analytical form and may require numerical methods or simulations to characterize. Therefore, whether it's possible to describe $p(y)$ depends on the characteristics of $h(x)$ and the distribution of x.

$$p(y) = \int_{-\infty}^{\infty} p(y, x) = \int_{-\infty}^{\infty} p(y|x)p(x)dx$$

where $p(y|x) \sim N(h(x), \sigma_r^2)$ because y = h(x) + r (more on this in b).

$$p(y) = \int_{-\infty}^{\infty} N(y; h(x), \sigma_r^2)p(x)dx = \int_{-\infty}^{\infty} p(x)\frac{1}{\sqrt{2\pi\sigma_r}} \exp\left(-\frac{(y - h(x))^2}{2\sigma_r^2}\right) dx$$

What we essentially can see from the previous equations is that p(y) depends on p(x) and p(h(x)). If we have linear transformation h(x) and p(x) will have the

same distribution and so will y also, however if h(x) is not linear it is hard to know the mixed distribution of P(y).

b)

It is possible to describe $p(y|x)$ since x is known and h(x) is deterministic, the value h(x) is basically constant, and y= constant + r. This means that the only distribution that is affecting y is r's distribution which is Gaussian distribution. Y is a Gaussian with the mean shifted

$$\mathcal{N}(h(x), \sigma_r^2)$$

The graphs for this case will basically look the same the linear normal gaussian case in e).

c)

In this case we have h(x) = Hx where H is a constant, this implies that h(x) is linear. So basically for p(y) we will have the same conclusion as in a) for linear case, the distribution of y will also be the same as the distribution of the transformation x. For $p(y|x)$ it is the exact same reasoning as in b) Y is a Gaussian with the mean shifted

$$\mathcal{N}(h(x), \sigma_r^2)$$

.

d)

For this case X is specifically Gaussian distributed, but our reasoning will still be the same as in a) and b) since they were generalised for all possible distribution of x. For $p(y|x)$ it is literally the same as b). For $p(y)$ we have two cases for when h(x) is linear or non linear. if h(x) is linear then y will be Gaussian distributed with the following

$$cov(y) = HCov(x)H^T + Cov(r)$$

$$\mu_y = H\mu_x$$

when we have the h(x) is not linear y is harder to estimate as explained in a). This means that y will not be Gaussian distributed and thus it is hard to estimate it.

e)

Simulations

We use two functions one linear and one non linear:

$$Linear : h(x) = 5x$$

$$Nonlinear : h(x) = 5sin(x)$$

We use two distributions for our simulation which are uniform and normal:

$$x \sim U(0, 10)$$

$$x \sim \mathcal{N}(0, 5)$$

In the figure we can strengthen our arguments from the above for different distributions. Below we can see when we have linear h(x) y will follow the distribution of x (one case it is uniform and one case it is normal).

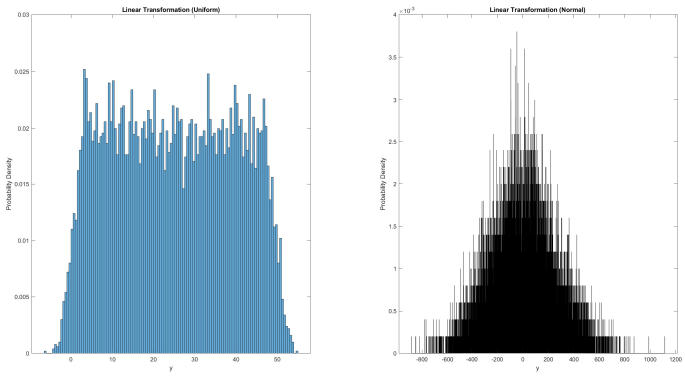

Figure 3.1: Linear transformation where h(x) = 5x

However, when h(x) is non linear y will have a weird distribution which explains why we can not describe p(y).

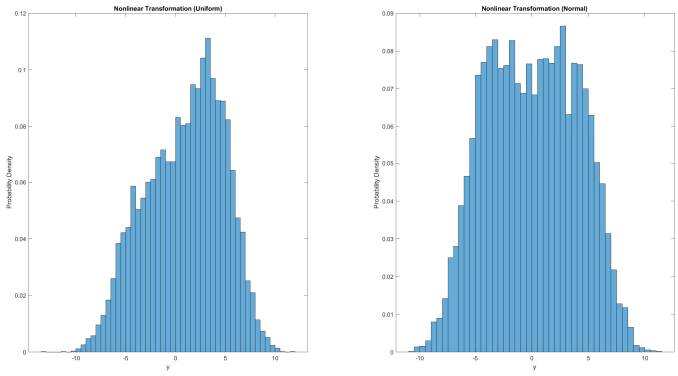

*Figure 3.2: Nonlinear transformation where h = 5 * sin(x)*

Now for the case of p(y|x) we assume x is known constant for example x=5 and simulate it. In the below figures we can see results which always seems to be a normal distribution with mean h(x) and and variance of r.

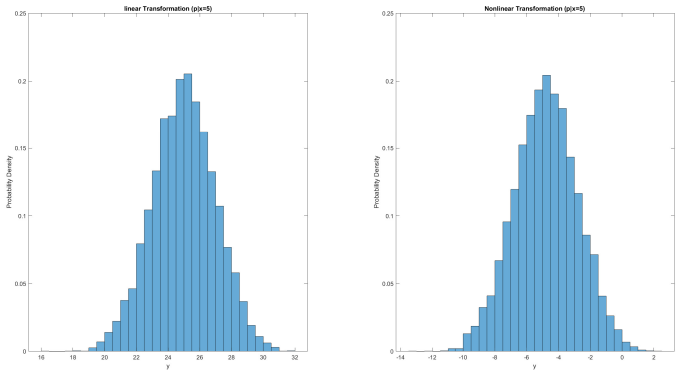

Figure 3.3: Nonlinear and linear transformation when x is given (assumed to be x=5)

For case of h(x)= Hx as mentioned this means that we will have a linear transformation, and if we assume H=5, the results for this will be the same as figure 3.1 for different distributions of x.

For the case of d) where x is Gaussian distribution it is already tackled since x were assumed to be gaussian as in the previous plots.

4 MMSE and MAP estimators

a)

The observed variable y follows a bimodal distribution because it is the sum of two random variables: θ and w. The random variable θ has a discrete distribution with two possible outcomes: -1 and 1, each occurring with equal probability. The random variable w follows a normal distribution $\mathcal{N}(0, \sigma^2)$, where $\sigma^2 = 0.5^2$ in this case.

When we add these two random variables to obtain $y = \theta + w$, we're essentially combining the discrete distribution of θ with the continuous distribution of w, resulting in a bimodal distribution for y.

The peaks in the histogram correspond to the modes of the bimodal distribution, with one peak around -1 and the other peak around $+1$, reflecting the contributions from the two modes of the discrete distribution of θ since θ can only take two values, -1 and 1.

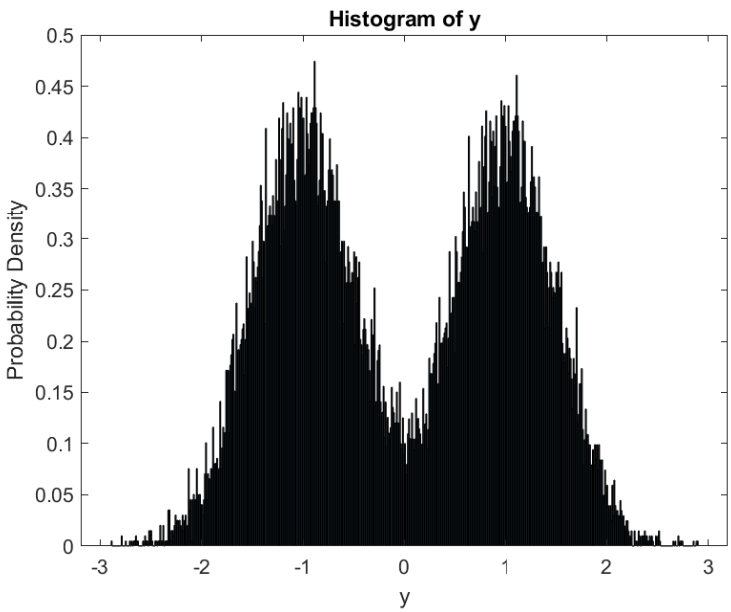

Figure 4.1: Bimodal

b)

The guess will be that theta is 1 because it is closest value y 0.7 and the variance is 0.5. Theta can be either 1 or -1. if theta is 1 then the range of y is between 1.5-0.5 and 0.7 is in this range.

We can prove this using Bayesian statistics

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

$$p(\theta = 1|y = 0.7) = \frac{p(y = 0.7|\theta = 1) * p(\theta = 1)}{p(y = 0.7)}$$

$$p(\theta = -1|y = 0.7) = \frac{p(y = 0.7|\theta = -1) * p(\theta = -1)}{p(y = 0.7)}$$

Now the probability that theta=1 or -1 is the same and since both equations have the same numerator, we only have to look at $p(y = 0.7|\theta = -1)$ and $p(y = 0.7|\theta = 1)$. $p(y|\theta)$ is a normal distribution $\mathcal{N}(\theta, 0.5^2)$. Now p(y=0.7) is higher when theta 1 than when -1, however there is still a possibility that it might be -1.

c)

To find p(y/theta) we say that theta is known and treat it like a constant. Which means that y will only depend on w and w had a Gaussian distribution. Y will be Gaussian distribution with mean θ and covariance σ^2. Now to find p(y)

$$p(y) = \int_{-\infty}^{\infty} p(y|\theta)p(\theta)d\theta$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2\sigma^2}} p(\theta)d\theta$$

$$= \frac{1}{\sigma\sqrt{2\pi}}(\int_{-\infty}^{\infty} 0.5 \cdot \delta(\theta - 1)e^{-\frac{(x-\theta)^2}{2\sigma^2}} d\theta + \int_{-\infty}^{\infty} 0.5 \cdot \delta(\theta + 1)e^{-\frac{(x-\theta)^2}{2\sigma^2}} d\theta)$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \cdot 0.5 \cdot e^{-\frac{(y+1)^2}{2\sigma^2}} + \frac{1}{\sigma\sqrt{2\pi}} \cdot 0.5 \cdot e^{-\frac{(y+1)^2}{2\sigma^2}}$$

d)

In this question we evaluate the posterior $P(y)$ by using $p(y)$, $p(\theta)$, $p(y \mid \theta)$, and Bayesian rule.

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

where:

$$p(y) = \frac{1}{\sigma\sqrt{2\pi}} \cdot 0.5 \cdot e^{-\frac{(y+1)^2}{2\sigma^2}} + \frac{1}{\sigma\sqrt{2\pi}} \cdot 0.5 \cdot e^{-\frac{(y+1)^2}{2\sigma^2}}$$

and

$$p(y|\theta) = \mathcal{N}(\theta, \sigma^2)$$

The results are written as follows

$$p(\theta = -1|y) = \frac{0.5 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y+1)^2}{2\sigma^2}}}{0.5 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-1)^2}{2\sigma^2}} + 0.5 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y+1)^2}{2\sigma^2}}} = \frac{1}{e^{\frac{2y}{\sigma^2}} + 1}$$

$$p(\theta = 1|y) = \frac{0.5 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-1)^2}{2\sigma^2}}}{0.5 \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-1)^2}{2\sigma^2}} + 0.5 \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y+1)^2}{2\sigma^2}}} = \frac{1}{e^{\frac{-2y}{\sigma^2}} + 1}$$

e)

In this question we should add the results from the previous tasks using the following

$$\hat{\theta}_{\text{MMSE}} = \sum_{\theta} \theta \Pr\{\theta|y\}$$

$$\hat{\theta}_{\text{MMSE}} = 1 \cdot p(\theta = 1|y) + (-1) \cdot p(\theta = -1|y) =$$

$$1 \cdot \frac{1}{e^{\frac{-2y}{\sigma^2}} + 1} + -1 \cdot \frac{1}{e^{\frac{2y}{\sigma^2}} + 1} = \frac{e^{\frac{2y}{a^2}} - 1}{e^{\frac{2y}{a^2}} + 1} = tanh\frac{y}{\sigma^2}$$

f)

In this question we should instead calculate the MAP estimator using

$$\theta_{\text{MAP}} = \arg\max_{\theta} p(\theta|y)$$

which is basically the estimation which maximizes the posterior. We should consider the two cases for when theta has different values. So the idea is basic if $p(\theta = 1|y)$ is larger than $p(\theta = -1|y)$ we should assume theta =1 and if it is the other way around we should assume theta = -1. Now using the following

$$p(\theta = 1|y) = \frac{1}{e^{\frac{-2y}{\sigma^2}} + 1}$$

$$p(\theta = -1|y) = \frac{1}{e^{\frac{2y}{\sigma^2}} + 1}$$

if y is larger or equal to 0 then $p(\theta = 1|y)$ is larger than $p(\theta = -1|y)$ which mean we should assume that $\theta_{\text{MAP}} = 1$. However when y is smaller than 0 then $p(\theta = -1|y)$ will be larger than $p(\theta = 1|y)$ and thus $\theta_{\text{MAP}} = -1$

g)

Now for the decision in b), it looks most like it would have been MAP estimator. However the MMSE tries to find a value which is based on how great the estimator performs on average which leads to it giving values which will not be viable since theta has only two possible values. But it will give a value close to 1 and we would have chosen 1. This would have been a great choice if we did not know values for theta. On the other hand, MAP finds the most probable choice using information that we already have such as what are the possible value theta can have and this is why this method was preferable used. A possible analogy is that MMSE tries to minimize its loses while MAP likes to take more risks which means it will win big when it guess if it guess correct but it might be more off when you it is wrong